

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Pseudocode Writing Task (Algorithm Design)

<i>Criteria</i>	Problem Understanding (2)	Algorithm Design (3)	Use of Control Structures (2)	Pseudocode Structure (1)	Completeness (2)	<i>Total(10)</i>
<i>Weightage</i>	20%	30%	20%	10%	20%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M.Mahathi(192324098)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M.Mahathi

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration. (BL3 – 1.05 Marks)

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:

Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. (BL4 – 1.05 Marks)

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:

Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3 – 1.05 Marks)*

Q4. Dataset: Student Course Enrollment**Student ID Courses Enrolled**

S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:

Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course "**Python**" are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4 – 1.05 Marks)*

Q5. Dataset: Hospital Treatment Hierarchy**Patient ID Treatments Received**

P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:

Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. Dataset : Online Shopping Transactions

order ID	Products Purchased
01	Mobile , Earphones
02	Mobile , Charger
03	Earphones , Charger
04	Mobile , Earphones , Charger
05	Mobile , Earphones

Question :

Design pseudocode for Apriori Algorithm to discover frequent product combinations.

Aim:- To find frequent product combinations from the online shopping transactions using a minimum support threshold.

Given :-

Assume Minimum Support = 60%.

Total transactions = 5

Minimum Support Count = $(5 \times 60) / 100 = 3$

Pseudocode

INPUT :

Transaction database D

Minimum support threshold minsup

OUTPUT:

Set of all frequent itemsets

BEGIN

$C_1 \leftarrow$ all unique products in D

FOR each itemset X in C_1 DO
 $\text{support_count}[X] \leftarrow$ number of transactions
 containing X

END FOR

$L_1 \leftarrow \{X \text{ in } C_1 \mid \text{support_count}[X] \geq \text{minsup}\}$

$K \leftarrow 2$

WHILE $L_{(K-1)}$ is not empty DO

$C_K \leftarrow \text{GenerateCandidates}(L_{(K-1)})$

 FOR each candidate X in C_K DO

 IF any $(K-1)$ -subset of X is NOT in $L_{(K-1)}$

 Remove X from C_K

 END IF

 END FOR

 FOR each candidate T in D DO

 FOR each candidate X in C_K DO

 IF X is a subset of T THEN

$\text{support_count}[X] \leftarrow \text{support}[X] + 1$

 END IF

 END FOR

 END FOR

$L_K \leftarrow \{X \text{ in } C_K \mid \text{support}[X] \geq \text{minsup}\}$

$K \leftarrow K + 1$

END WHILE

RETURN $L_1 \cup L_2 \cup \dots \cup L_{(K-1)}$

END

Q2: FP-Growth Library Book Borrowing

DataSet

Borrow ID	Book
B ₁	Data Science, Python
B ₂	Python, Machine Learning
B ₃	Data Science, Machine Learning
B ₄	Data Science, python, Machine Learning
B ₅	Python, Machine learning

Minimum Support Count = 2

Pseudocode

FP-GROWTH (D, min-Support)

1. Scan D and calculate support of each item.
2. Remove items whose support < min-support.
3. Sort remaining items according to decreasing support
4. Create an empty F-P Tree
5. For each transaction items T in D:
Remove infrequent items
Sort-items T by frequency
update node count.
6. Create header table linking identical item
7. Mine the FP-Tree recursively:
Find conditional pattern bases.
construct conditional FP-Trees
Generate frequency item sets.
8. RETURN are frequent itemsets

Q4: Constraint - Based Frequent item transactions

Q3. Association rule Generation

Dataset

Itemset	Support Count
{Internet}	4
{streaming}	4
{Cloud storage}	3
{Internet, streaming}	3
{streaming, cloud storage}	2

Assume minimum Confidence = 60%.

$$\text{Confidence } (A \rightarrow B) = \frac{\text{Support } (A \cup B)}{\text{Support } (A)} \times 100$$

Pseudocode

GENERATE - RULES (FREQUENT - Items, min-confidence)

1. For each frequent itemset F;
2. Generate all non-empty subsets A of F
3. $B \in F - A$
4. Confidence $\leftarrow \text{Support } (F) / \text{Support } (A)$
5. IF confidence $\geq$ min-confidence;
6. Generate rule $A \rightarrow B$
7. Accept the rule
8. ELSE:
9. Reject the rule
10. RETURN all accepted rule

Q4: Constraint — Based Frequent item set Mining
DataSet

Student	Courses
S1	python , Database
S2	python , AI
S3	Database , AI
S4	python , Database , AI
S5	python

Final frequent itemsets

{python}

{python , Database}

{python , AI}

Pseudocode

1. Define constraint

Every itemset must contain "python".

2. Generate c_1

3. Remove candidate that do not contain python

4. Count support

5. $L \leftarrow$ candidates satisfying min-support

6. FOR $k=2$ while $L(k-1)$ is not empty :

Generate candidates itemset from $L(k-1)$

For each candidate c :

If "python" is not in c

prune c

ELSE

Add c to L_k

7. RETURN all L_k

5: Multi-level Association Rule Mining

Dataset

Patient	Treatments
P1	Healthcare, cardiology, ECG
P2	Healthcare, orthopedics
P3	Healthcare, cardiology
P4	Healthcare, cardiology, ECG
P5	Healthcare, orthopedics

Pseudocode

MULTILEVEL - ASSOCIATION - MINING (D, Hierarchy, min-support)

1. Define the treatment hierarchy
2. Start from the highest (general) level
3. For each hierarchy level:
 - Extract items belonging to that level
 - Count support of each item (item set)
 - Remove items below min-support
 - Generate association rule from frequency items.
 - Calculate confidence for each rule
 - Keep rules satisfying min-confidence
4. Move to the next, more detailed hierarchy level.
5. Repeat the mining process.
6. RETURN frequent patterns & rule from all levels.